

MERN STACK

DAY – 16

13 July 2026

REACT

React

React is a JavaScript library for building user interfaces (UI). It helps create fast, reusable, and interactive web applications using components.

Features

- Component-based architecture
- Virtual DOM for better performance
- Reusable UI components
- One-way data flow
- JSX syntax

```
function App() {  
  
  return (  
  
    <h1>Hello, React!</h1>  
  
  );  
  
}  
  
export default App;
```

React App Setup

1. Install Node.js

Download and install Node.js from the official website.

Check installation:

```
node -v  
  
npm -v
```

2. Create a React App

`npx create-react-app` is a command used to create a new React project with all the required files and dependencies automatically.

Syntax

```
npx create-react-app my-app
```

- npx → Runs a package without installing it globally.
- create-react-app → Tool that sets up a React application.
- my-app → The name of your project folder (you can replace it with any name).

React Project Structure

A React project structure is the organized arrangement of folders and files in a React application. It provides a clear and systematic way to organize the application's source code, components, assets, styles, and dependencies. A standard React project typically includes folders such as `src`, `public`, and `node_modules`, along with files like `package.json`, `App.js/App.jsx`, and `index.js/main.jsx`. The `src` folder contains the application's main source code, reusable components, CSS files, and assets, while the `public` folder stores static files such as the HTML template and images. The `node_modules` folder contains all the installed packages and libraries required by the project. A well-organized project structure improves code readability, simplifies maintenance, encourages code reusability, and makes the application easier to scale as it grows.

Components

A React component is a reusable, independent, and self-contained piece of the user interface that represents a specific part of a web application. Components allow developers to divide a large application into smaller, manageable sections such as a header, navigation bar, content area, footer, or button. Each component contains its own logic and UI, making it easy to reuse across different parts of the application. React primarily uses functional components, which are simple JavaScript functions that return JSX and can use Hooks for managing state and other features. By using components, developers can write cleaner, more organized, and maintainable code, reduce duplication, and build complex user interfaces efficiently.

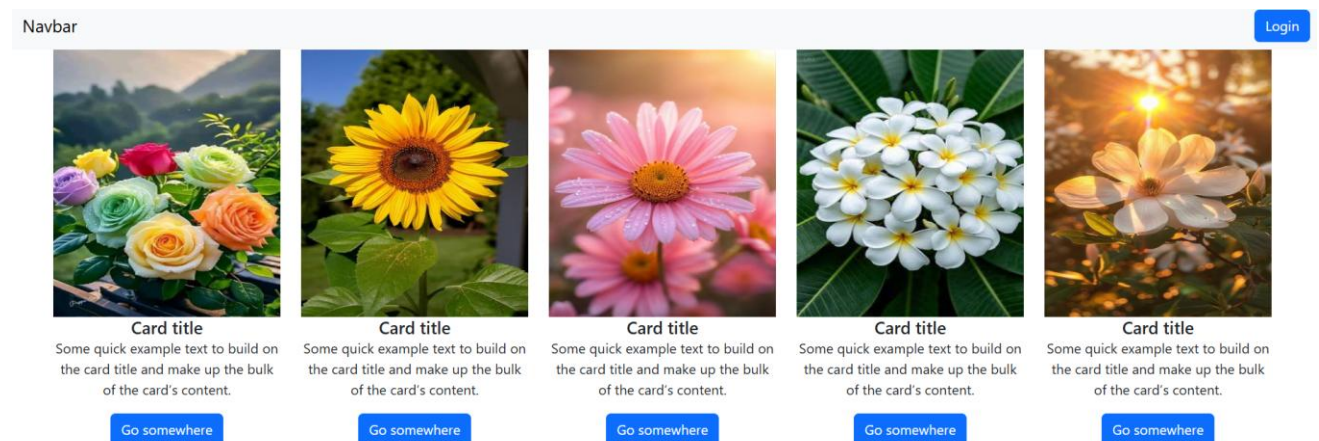
Import

- Import is used to bring components, functions, variables, or modules from one file into another.
- It allows different files to use reusable code.
- The import statement is usually written at the top of a file.
- It helps organize React applications into multiple files.
- It improves code readability and maintainability.

Export

- Export is used to make a component, function, variable, or module available for use in other files.
- It enables code sharing between different files.
- React supports two types of exports:
 - Default Export – Only one default export is allowed per file.
 - Named Export – Multiple named exports are allowed in the same file.
- Exporting components makes them reusable throughout the application.

TASK



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